

LucyTrace: Installation & Setup Guide ☒

This document provides a quick reference for installing the necessary tools to run LucyTrace on **Windows**, **macOS**, and **Linux**.

☒ Prerequisites

LucyTrace is a script that orchestrates three powerful engines. You need all of them installed and accessible in your system `PATH`:

1. **Python 3.x** (The conductor)
 2. **ImageMagick** (The cleaner)
 3. **Potrace** (The tracer)
 4. **Inkscape** (The exporter)
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☒ Windows Installation

1. Python 3.x

1. Download from python.org.
2. **IMPORTANT**: During installation, check the box **"Add Python to PATH"**.
3. Verify in PowerShell: `python --version`

2. ImageMagick

1. Download the **DLL installer** from [ImageMagick.org](https://imagemagick.org).
2. During installation, check **"Install legacy utilities (e.g. convert)"** and **"Add to PATH"**.

3. Potrace

1. Download the **Windows (64-bit)** zip from [SourceForge](https://sourceforge.net/projects/potrace).
2. Extract the zip. You will see `potrace.exe`.
3. Move the extracted folder to a permanent spot (e.g., `C:\Program Files\potrace`).
4. **Add to PATH**:
 - Press `Win` Key, type "env", and select **"Edit the system environment variables"**.
 - Click **Environment Variables** -> System Variables -> Select `Path` -> Edit.
 - Click **New** and paste the folder path (e.g., `C:\Program Files\potrace`).
5. Verify: `potrace --version`

4. Inkscape

1. Download installer from inkscape.org.
 2. Install and ensure **"Add to PATH"** is selected.
 3. Verify: `inkscape --version`
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☒ macOS Installation

1. Python 3.x

macOS 12+ usually has Python, but Homebrew is recommended:

```
brew install python
python3 --version
```

2. ImageMagick

```
brew install imagemagick
```

3. Potrace

```
brew install potrace
```

4. Inkscape

1. Install via Homebrew: `brew install --cask inkscape`
2. **CRITICAL STEP:** macOS often hides the command line tool inside the App bundle. Link it manually:

```
sudo ln -s /Applications/Inkscape.app/Contents/MacOS/inkscape /usr/local/bin/inkscape
```

3. Verify: `inkscape --version`

Linux Installation (Debian/Ubuntu)

```
sudo apt update
sudo apt install python3 python3-pip imagemagick potrace inkscape
```

Optional: Virtual Environment (venv)

Using a virtual environment prevents Python version conflicts.

1. Navigate to the LucyTrace folder:

```
cd /path/to/LucyTrace
```

2. Create the venv:

```
# Windows
python -m venv venv

# Mac/Linux
python3 -m venv venv
```

3. **That's it!** The `lucytrace.bat` (Windows) and `lucytrace.sh` (Mac) wrappers automatically detect the `venv` folder and use it if it exists.

Final Verification

Open your terminal or command prompt and run these four commands. If any fail, the pipeline will not work.

```
python --version    # (or python3 --version)
magick -version
potrace --version
inkscape --version
```